

Task 81: Coin Change

PROBLEM DESCRIPTION

You are given an integer array `coins` representing coins of different denominations and an integer `amount` representing a total amount of money. Return the fewest number of coins that you need to make up that amount. If that amount of money cannot be made up by any combination of the coins, return -1.

JAVA SOLUTION

```
import java.util.Arrays;

class Solution {
    public int coinChange(int[] coins, int amount) {
        int[] dp = new int[amount + 1];
        Arrays.fill(dp, amount + 1);
        dp[0] = 0;

        for (int i = 1; i <= amount; i++) {
            for (int coin : coins) {
                if (coin <= i) {
                    dp[i] = Math.min(dp[i], dp[i - coin] + 1);
                }
            }
        }

        return dp[amount] > amount ? -1 : dp[amount];
    }
}
```